Soft Condensed Matter

A Comprehensive Course Summary

Part II Physics — University of Cambridge

A self-contained tour of the whole course: fluid dynamics at low Reynolds number, viscoelasticity and Brownian motion, polymer physics, surfaces and wetting, colloids and charged interfaces, and self-assembly. Each chapter gives the key concepts, the core equations with every symbol defined, the essential results and how they are obtained, and a closing box of the ideas and pitfalls that matter most. Notation follows the course lecture notes.

Fluid Dynamics · **Viscoelasticity & Brownian Motion** · **Polymers** · **Surfaces & Wetting** · **Colloids & Electrostatics** · **Self-Assembly**

The unifying theme: *soft* matter is governed by energies comparable to $k_{\mathrm{B}}T$, so thermal fluctuations, entropy, and weak interactions — not strong chemical bonds — set the physics.

Contents

1 Orientation: What Makes Matter "Soft"

The defining idea

Soft condensed matter — polymers, colloids, surfactants, gels, liquid crystals, biological materials — is characterised by structures whose characteristic energy scales are comparable to the thermal energy $k_B T$ at room temperature ($\approx 4 \times 10^{-21}$ J, or ≈ 25 meV). As a result these materials are easily deformed, dominated by *entropy* as much as energy, and perpetually agitated by thermal (Brownian) motion. The same $k_B T$ that would be a negligible perturbation to a hard crystal is the protagonist here.

Three consequences run through the entire course:

- **Entropy is a force.** Stretching a polymer, confining a membrane, or mixing two liquids all change the number of accessible microstates, and the resulting entropic free-energy changes produce real, measurable forces (the entropic spring, steric stabilisation, the osmotic pressure of mixing).
- **Thermal fluctuations are always present.** Colloids diffuse, membranes undulate, interfaces ripple — all with amplitudes set by equipartition ($\frac{1}{2}k_B T$ per mode).
- **Dynamics are overdamped.** At the small scales and low speeds of soft matter, the Reynolds number is tiny: inertia is negligible and motion is dominated by viscous drag. "Coasting" does not happen; remove the force and motion stops essentially instantly.

The course is organised into six chapters, summarised in turn below: fluid dynamics (the overdamped, low-Reynolds world), viscoelasticity and Brownian motion (how soft materials respond to stress and to thermal noise), polymers (the statistical mechanics of long chains), surfaces and wetting (the physics of interfaces), colloids and charged interfaces (what keeps suspensions stable), and self-assembly (how amphiphiles and membranes organise themselves).

2 Fluid Dynamics at Low Reynolds Number

2.1 Key Concepts

- **Navier–Stokes equation:** Newton's second law for a fluid element, balancing inertia against pressure, viscous and external forces.
- **Reynolds number Re:** the dimensionless ratio of inertial to viscous forces. Soft matter lives at $Re \ll 1$, where inertia is negligible.
- **Stokes (creeping) flow:** the linear, time-reversible flow obtained when the nonlinear inertial term is dropped at low Re.
- **Kinematic reversibility & the scallop theorem:** at $Re \ll 1$ a reciprocal (time-symmetric) stroke produces no net motion, so microorganisms must swim with non-reciprocal strokes (rotating flagella, travelling waves).
- **Momentum diffusion:** viscosity makes transverse momentum diffuse through a fluid with diffusivity $\nu = \eta/\rho$ (the kinematic viscosity).
- **Stokes drag:** the $F = 6\pi\eta a v$ friction on a sphere, the single most-used result of the chapter.
- **Poiseuille flow & hydraulic resistance:** pressure-driven flow in channels, with resistance $\propto 1/R^4$ for a pipe.

2.2 Core Equations

Navier–Stokes and the Reynolds number

$$\rho[\partial_t \vec{v} + (\vec{v}\cdot\nabla)\vec{v}] = -\nabla p + \eta\nabla^2\vec{v} + \vec{f}, \qquad Re = \frac{\rho v_0 L_0}{\eta}.$$

ρ is the fluid density ($\mathrm{kg\,m^{-3}}$), $\vec{v}$ the velocity field ($\mathrm{m\,s^{-1}}$), p the pressure (Pa), η the dynamic viscosity (Pa s), $\vec{f}$ a body-force density ($\mathrm{N\,m^{-3}}$); v_0, L_0 are characteristic velocity and length scales. Non-dimensionalising shows the inertial terms carry a prefactor Re: when $Re \ll 1$ they drop out, leaving the linear **Stokes equation** $0 = -\nabla p + \eta\nabla^2\vec{v}$.

Stokes drag and Poiseuille flow

$$F = 6\pi\eta a v, \qquad Q = \frac{\pi R^4}{8\eta L}\Delta p, \qquad R_{\text{hyd}} = \frac{8\eta L}{\pi R^4}.$$

a is the sphere radius, R the pipe radius, L its length, Δp the pressure drop, and Q the volumetric flow rate ($\mathrm{m^3\,s^{-1}}$).

2.3 Essential Results and How They Arise

Why Re controls everything. Rescaling all lengths by L_0 and velocities by v_0 turns Navier–Stokes into $Re[\tilde{\partial}_t\tilde{\vec{v}} + (\tilde{\vec{v}}\cdot\tilde{\nabla})\tilde{\vec{v}}] = -\tilde{\nabla}\tilde{p} + \tilde{\nabla}^2\tilde{\vec{v}}$. A single dimensionless group Re multiplies the entire (nonlinear) inertial side. For a 1 μm colloid moving at $\mu\mathrm{m\,s^{-1}}$ in water, $Re \sim 10^{-6}$ — inertia is utterly negligible.

Stokes drag by momentum diffusion. A moving sphere is a steady source of momentum; in steady Stokes flow the transverse velocity field decays as $v_T(r) = v\,a/r$. The viscous stress $\sim \eta v a/r^2$ integrated over the surface area $4\pi a^2$ gives $F \approx 4\pi\eta a v$; the exact solution replaces 4π by 6π. The lesson is that drag scales as the radius a, not the area a^2 — because momentum *diffuses* away rather than being swept up.

Poiseuille flow. A pressure gradient balanced by viscous shear ($\eta\,\partial_z^2 v = -\Delta p/L$) integrates, with no-slip walls, to a parabolic velocity profile; integrating the profile across a circular pipe gives $Q \propto R^4$. This dramatic R^4 dependence is why narrow channels resist flow so strongly and why the electrical-circuit analogy (series/parallel hydraulic resistances) is so useful.

2.4 Diagram Analysis

The slides show flow past a sphere at increasing Re: smooth, fore–aft symmetric streamlines at $Re \ll 1$ that progressively develop a wake, recirculation, and finally turbulent vortex shedding at high Re. The fore–aft symmetry at low Re is the visual signature of *time-reversibility*: the Stokes equation has no time derivative, so reversing the driving exactly reverses the flow. This is the origin of the scallop theorem.

Key ideas & pitfalls

1. "Low Reynolds number means slow." Not quite — it means viscous forces dominate inertial ones, which can happen through small size, low speed, or high viscosity. A bacterium and a ball-bearing in honey both live at low Re.
2. Forgetting reversibility. At $Re \ll 1$ a time-symmetric deformation produces zero net displacement. This is deeply counter-intuitive (a scallop opening and closing goes nowhere) and a favourite exam point.
Takeaway: the whole chapter flows from one move — drop the Re-weighted inertial term — which linearises the dynamics and makes drag, channel flow, and swimming all tractable.

3 Viscoelasticity and Brownian Motion

3.1 Key Concepts

- **Strain and stress:** dimensionless deformation ε and force-per-area σ; linear elasticity relates them by Hooke's law with moduli E (Young), G (shear), K (bulk), and Poisson ratio ν.
- **Viscoelasticity:** materials that are part-solid (elastic, store energy) and part-liquid (viscous, dissipate energy); their response depends on timescale.
- **Complex modulus $G^*(\omega) = G' + iG''$:** the storage modulus G' (elastic, in-phase) and loss modulus G'' (viscous, out-of-phase) under oscillatory strain.
- **Maxwell, Kelvin–Voigt, Zener models:** spring-and-dashpot caricatures capturing stress relaxation, creep, or both.
- **Brownian motion & the Langevin equation:** the stochastic equation of motion for a particle buffeted by solvent collisions.
- **Fluctuation–dissipation theorem:** the same friction that dissipates energy sets the strength of the random force.
- **Einstein & Stokes–Einstein relations:** diffusion constant $D = k_B T/\gamma$, and $D = k_B T/6\pi\eta r$ for a sphere.
- **Kramers escape:** thermally activated barrier crossing, giving Arrhenius-like rates.

3.2 Core Equations

Viscoelastic building blocks

$$\text{Maxwell (series): } \dot{\varepsilon} = \frac{\dot{\sigma}}{G_0} + \frac{\sigma}{\eta}, \qquad \text{Kelvin–Voigt (parallel): } \sigma = G\varepsilon + \eta\dot{\varepsilon}.$$

σ is stress (Pa), ε strain (dimensionless), G_0, G shear moduli (Pa), η viscosity (Pa s); the relaxation time is $\tau = \eta/G$.

Langevin equation and its consequences

$$m\dot{\vec{v}} = -\gamma\vec{v} + \vec{\xi}(t), \quad \left\langle \vec{\xi}(t)\cdot\vec{\xi}(t') \right\rangle = 2\gamma k_B T\,\delta(t-t') \times (\text{dim}),$$

$$\left\langle \vec{x}^2 \right\rangle = 6Dt, \qquad D = \frac{k_B T}{\gamma} = \frac{k_B T}{6\pi\eta r}.$$

m is the particle mass, $\gamma = 6\pi\eta r$ the drag coefficient (kg s^{-1}), $\vec{\xi}$ the random force, D the diffusion coefficient (m^2 s^{-1}), r the particle radius.

3.3 Essential Results and How They Arise

Storage and loss moduli. Under oscillatory strain $\varepsilon = \varepsilon_0 e^{i\omega t}$, the stress response defines $G^*(\omega) = \sigma/\varepsilon$. For the Maxwell model $G^* = i\omega G_0\eta/(G_0 + i\omega\eta)$: at high frequency $G^* \to G_0$ (solid-like, elastic) and at low frequency $G^* \to i\omega\eta$ (liquid-like, viscous). The crossover at $\omega\tau = 1$ is the material's characteristic relaxation.

Which model does what. A Maxwell element (series) relaxes stress exponentially ($G(t) = G_0 e^{-t/\tau}$) but flows without bound under constant load — good for stress relaxation, useless for creep. A Kelvin–Voigt element (parallel) creeps to a bounded strain $\varepsilon = (\sigma_0/G)(1 - e^{-t/\tau})$ but cannot relax stress at all (holding strain fixed freezes the dashpot). The Zener (standard linear solid) combines them and captures *both* bounded creep and finite stress relaxation — the minimal realistic model.

Fluctuation–dissipation. Solving the Langevin equation for the velocity autocorrelation and imposing equipartition ($\frac{1}{2}m\left\langle v^2 \right\rangle = \frac{3}{2}k_B T$) fixes the noise strength to $2\gamma k_B T$ per dimension. Integrating to long times gives $\left\langle x^2 \right\rangle = 6Dt$ with $D = k_B T/\gamma$ (Einstein), and inserting Stokes drag gives Stokes–Einstein $D = k_B T/6\pi\eta r$. The profound content: drag and noise are two faces of the same molecular collisions.

Kramers escape. A particle in a potential well of barrier height ΔE escapes at a thermally activated rate $k \sim \nu_0\, e^{-\Delta E/k_B T}$. This Arrhenius form governs reaction kinetics, nucleation, and conformational transitions throughout soft matter.

3.4 Diagram Analysis

The spring–dashpot diagrams encode the series/parallel logic visually: elements in *series* share stress and add strains (Maxwell); elements in *parallel* share strain and add stresses (Kelvin–Voigt). The $G'(\omega)$, $G''(\omega)$ plots show G' plateauing at high frequency (elastic solid) while G'' peaks near $\omega\tau = 1$ (maximum dissipation at the relaxation frequency) — the standard rheological fingerprint.

Key ideas & pitfalls

1. Series vs parallel. The single most common confusion. Series $\Rightarrow$ equal stress, strains add. Parallel $\Rightarrow$ equal strain, stresses add. Get this right and the model behaviours follow.
2. Why a single element is never enough. Each two-element model has one timescale and gets only one of {creep, relaxation} right. Real materials need the three-element Zener model (or a spectrum of relaxation times).
Takeaway: Brownian motion is the microscopic root of viscous drag; the fluctuation–dissipation theorem ties the random kicks to the systematic friction, and Stokes–Einstein turns a tracked trajectory into a molecular ruler.

4 Polymers

4.1 Key Concepts

- **Ideal chain:** a random walk of N uncorrelated segments; end-to-end distance scales as $\langle R^2 \rangle = N b_0^2$, i.e. $R \sim N^{1/2}$.
- **Models of the ideal chain:** freely jointed, freely rotating, and bead–spring (Gaussian) chains — all give $R \sim N^{1/2}$ at large N.
- **Kuhn length & persistence length:** the effective segment length b and the decay length l_p of bond-orientation correlations; they set the scale at which the chain looks flexible.
- **Entropic elasticity:** a stretched ideal chain resists with a purely entropic spring force $\propto k_{\mathrm{B}}T$.
- **Excluded volume & real chains:** self-avoidance swells the chain to $R \sim N^{3/5}$ (Flory exponent $\nu \approx 3/5$) in a good solvent.
- **Coil–globule transition:** the chain collapses from a swollen coil to a dense globule as solvent quality worsens.
- **Flory–Huggins theory:** lattice free energy of mixing governing polymer solubility and phase separation; the critical point $\chi_c \approx \frac{1}{2} + 1/\sqrt{N}$.
- **Rubber elasticity:** cross-linked networks of entropic springs, modulus $G = \nu k_{\mathrm{B}}T$.
- **Reptation:** the snake-like motion of a chain confined in a mesh or melt, giving length-dependent dynamics.

4.2 Core Equations

Chain statistics and entropic spring

$$\langle R^2 \rangle = N b_0^2, \qquad F(\vec{R}) = \frac{3k_{\mathrm{B}}T}{2Nb_0^2}\vec{R}^2 + \text{const}, \qquad \vec{f} = \frac{3k_{\mathrm{B}}T}{Nb_0^2}\vec{R}.$$

N is the number of segments, b_0 the segment length (m), $\vec{R}$ the end-to-end vector. The chain behaves as a Hookean spring of constant $k = 3k_{\mathrm{B}}T/Nb_0^2$ — stiffer when hotter, the hallmark of entropy.

Flory–Huggins and the critical point

$$\frac{f_{\mathrm{mix}}}{k_{\mathrm{B}}T} = \frac{\Phi}{N}\ln\Phi + (1-\Phi)\ln(1-\Phi) + \chi\,\Phi(1-\Phi), \qquad \Phi_c = \frac{1}{\sqrt{N}+1}, \qquad \chi_c \approx \frac{1}{2} + \frac{1}{\sqrt{N}}.$$

Φ is the polymer volume fraction (dimensionless), χ the Flory interaction parameter (dimensionless), N the degree of polymerisation.

Real-chain scaling and rubber modulus

$$\langle R^2 \rangle \sim b^2 N^{6/5} \;\; (\nu \approx 3/5,\ \text{good solvent}), \qquad G = \nu_s k_{\mathrm{B}}T, \qquad \sigma = \nu_s k_{\mathrm{B}}T\left(\lambda - \frac{1}{\lambda^2}\right).$$

ν_s is the number density of network strands (m^{-3}), λ the stretch ratio.

4.3 Essential Results and How They Arise

Random-walk statistics. Summing N independent bond vectors and using $\langle \vec{u}_i \cdot \vec{u}_j \rangle = 0$ for $i \neq j$ gives $\langle R^2 \rangle = Nb_0^2$: the chain size grows only as $\sqrt{N}$, far below the contour length Nb_0, because the walk doubles back on itself. The central limit theorem makes the end-to-end vector Gaussian-distributed.

Entropic spring. The entropy of a chain pinned at end-to-end vector $\vec{R}$ is $S = k_\mathrm{B} \ln P(\vec{R})$, quadratic in R; with no internal energy, $F = -TS = \frac{3k_\mathrm{B}T}{2Nb_0^2}R^2$. The restoring force is linear in R and proportional to temperature — heating a stretched rubber band makes it pull harder.

Excluded volume. Real segments cannot overlap, so the chain swells. Flory's argument balances the entropic cost of stretching ($\sim R^2/Nb^2$) against the energetic cost of self-overlap ($\sim N^2/R^3$); minimising gives $R \sim bN^{3/5}$ in 3D. In a poor solvent the balance reverses and the chain collapses to a dense globule ($R \sim N^{1/3}$); the crossover is the coil–globule transition, with the ideal $\nu = 1/2$ recovered at the theta point $\chi = \frac{1}{2}$.

Flory–Huggins. Lattice counting gives an entropy of mixing (reduced by $1/N$ for the polymer, since its segments move together) plus a mean-field contact energy $\chi\Phi(1-\Phi)$. The free-energy curvature sets the spinodal; its extremum is the critical point. The factor $1/N$ is why long polymers phase-separate so readily ($\chi_c \to \frac{1}{2}$ and $\Phi_c \to 0$ as $N \to \infty$).

Rubber elasticity. Treating each network strand as an entropic spring deforming affinely with the bulk, the free energy of an incompressible uniaxial stretch is $\Delta F = \frac{nk_\mathrm{B}T}{2}(\lambda^2 + 2/\lambda - 3)$, giving the nonlinear stress $\sigma = \nu_s k_\mathrm{B}T(\lambda - \lambda^{-2})$ and modulus $G = \nu_s k_\mathrm{B}T$.

4.4 Diagram Analysis

The random-walk sketches show how a chain conformation maps onto a walk on a lattice, making the Gaussian end-to-end distribution intuitive. The force–extension curves (e.g. the freely-jointed-chain tanh law) show a linear Hookean regime at low force crossing over to a plateau at the contour length — the chain cannot be stretched beyond fully extended. The Flory–Huggins phase diagrams plot χ (or $1/T$) against Φ, with the binodal enclosing the two-phase region and the spinodal inside it, meeting at the critical point; the asymmetry (critical point at small Φ) is the polymer signature.

Key ideas & pitfalls

1. R is not the contour length. The chain size $\sim N^{1/2}b$ (ideal) or $N^{3/5}b$ (swollen) is far smaller than the contour length Nb. Confusing the two is a classic error.
2. The entropic spring stiffens with temperature. Opposite to a bond (energetic) spring. This counter-intuitive sign is the essence of rubber elasticity.
3. The $1/N$ in Flory–Huggins entropy. It applies only to the polymer term (segments are tied together), not the solvent term. Dropping or misplacing it changes the critical point qualitatively.
Takeaway: polymer physics is the statistical mechanics of long chains — entropy dominates, and a handful of scaling laws ($N^{1/2}$, $N^{3/5}$, $\chi_c \approx \frac{1}{2}$) organise the whole subject.

5 Surfaces, Interfaces and Wetting

5.1 Key Concepts

- **Surface tension / surface energy γ:** the free-energy cost per unit area of an interface; equivalently a force per unit length.

- **Laplace pressure:** the pressure jump across a curved interface, $\Delta p = \gamma(1/R_1 + 1/R_2)$.
- **Capillary rise (Jurin's law):** the height a wetting liquid climbs in a thin tube.
- **Capillary length $\ell_c = \sqrt{\gamma/\rho g}$:** the scale below which surface tension dominates gravity (≈ 2.7 mm for water).
- **Young's equation:** the contact-angle force balance at a three-phase line.
- **Wetting vs dewetting:** set by the spreading parameter S.
- **Gibbs adsorption isotherm:** how surfactants lower surface tension, and how this defines the CMC.
- **Cahn–Hilliard interface:** the diffuse-interface description balancing gradient energy against a double-well potential.

5.2 Core Equations

Surface tension results

$$\Delta p = \gamma\left(\frac{1}{R_1} + \frac{1}{R_2}\right), \qquad h = \frac{2\gamma\cos\Theta}{\rho g R} \text{ (Jurin)}, \qquad \gamma_{SL} + \gamma_{LV}\cos\Theta = \gamma_{SV} \text{ (Young)}.$$

γ is the surface tension ($\mathrm{N\,m^{-1}}$ or $\mathrm{J\,m^{-2}}$), R_1, R_2 the principal radii of curvature, Θ the contact angle, ρ density, g gravity, R the capillary radius; $\gamma_{SL}, \gamma_{LV}, \gamma_{SV}$ are the three interfacial tensions.

Spreading parameter and Gibbs isotherm

$$S = \gamma_{SV} - (\gamma_{SL} + \gamma_{LV}), \qquad \frac{n}{A} = -\frac{1}{k_B T}\frac{\mathrm{d}\gamma}{\mathrm{d}\ln c}.$$

$S > 0$ gives total wetting; n/A is the surface excess of solute per unit area, c the bulk concentration.

5.3 Essential Results and How They Arise

Surface tension's origin. Molecules at an interface have fewer favourable neighbours than in the bulk, so creating area costs energy; this energy-per-area is γ. Mechanically it acts as a line tension pulling to minimise area — which is why small droplets are spherical.

Laplace and Jurin. Minimising surface area under a volume constraint shows a curved interface supports a pressure jump $\Delta p = \gamma(1/R_1 + 1/R_2)$. In a capillary the wetting meniscus curvature drives liquid up until the hydrostatic head $\rho g h$ balances the Laplace pressure, giving $h = 2\gamma\cos\Theta/\rho g R$ — *thinner tubes lift liquid higher.*

Young's equation. Balancing the three interfacial tensions horizontally at the contact line gives $\gamma_{SV} = \gamma_{SL} + \gamma_{LV}\cos\Theta$. When the spreading parameter $S > 0$ there is no equilibrium angle and the liquid spreads completely (total wetting).

Gibbs isotherm and the CMC. Surfactant adsorbs at the liquid–vapour interface, lowering γ as bulk concentration rises; the slope $\mathrm{d}\gamma/\mathrm{d}\ln c$ gives the surface coverage. Above the critical micelle concentration the monomer activity saturates, so γ stops falling — the kink in γ vs $\ln c$ *is* the CMC.

5.4 Diagram Analysis

The droplet sketches show how the contact angle grows from 0° (complete wetting, hydrophilic) through 90° to > 90° (hydrophobic) and toward 180° (complete dewetting, e.g. water on a lotus leaf), tracking the sign and magnitude of $\cos\Theta$. The γ-versus-$\ln c$ curves for surfactant, alcohol and salt contrast a sharp CMC kink (surfactant), a gradual decrease (weak amphiphile), and a slight *increase* (salt, which is repelled from the interface).

Key ideas & pitfalls

1. Surface tension is energy per area *and* force per length. The two units ($\mathrm{J\,m^{-2}} = \mathrm{N\,m^{-1}}$) are identical; use whichever picture (energy minimisation or force balance) fits the problem.
2. Young's equation is a horizontal balance. The vertical component of γ_{LV} is taken up by elastic deformation of the solid and is usually ignored.
Takeaway: every interfacial phenomenon — droplet shape, capillary rise, wetting, surfactant action — flows from one quantity, the area cost γ, applied either as an energy or as a tension.

6 Colloids and Charged Interfaces

6.1 Key Concepts

- **Colloid:** a dispersion of particles (∼nm–μm) in a medium, large enough to scatter light but small enough to be kept suspended by Brownian motion.
- **Van der Waals / dispersion attraction:** the always-present $\sim -1/r^6$ atomic attraction, summed to a $-A/12\pi h^2$ plate–plate form via the Hamaker constant A.
- **Electric double layer:** the cloud of counter-ions screening a charged surface.
- **Bjerrum length l_B:** the separation at which two elementary charges interact with energy $k_\mathrm{B}T$ (≈ 0.7 nm in water).
- **Poisson–Boltzmann & Debye–Hückel:** the equations for the double-layer potential; linearisation gives exponential screening over the Debye length λ_D.
- **Debye screening length λ_D:** the range of the screened electrostatic repulsion, tunable by salt.
- **DLVO theory:** the sum of van der Waals attraction and screened-Coulomb repulsion, predicting colloidal stability.
- **Steric stabilisation:** polymer brushes providing salt-independent repulsion (the alternative to charge stabilisation).

6.2 Core Equations

Characteristic lengths and interactions

$$l_B = \frac{e^2}{4\pi\epsilon_0\epsilon_r k_\mathrm{B}T}, \qquad \lambda_D = \kappa^{-1}, \qquad \kappa^2 = \frac{e^2}{\epsilon k_\mathrm{B}T}\sum_i z_i^2 n_{i0},$$

$$U_\mathrm{vdW}(h) = -\frac{A}{12\pi h^2}, \qquad U_e(h) \propto \mathrm{e}^{-\kappa h} \text{ (screened Coulomb).}$$

l_B is the Bjerrum length (m), λ_D the Debye length (m), κ the inverse screening length, n_{i0} the bulk number density of ion species i, z_i its valency, A the Hamaker constant (J), h the surface separation (m).

DLVO potential

$$U_\mathrm{DLVO}(h) = U_e(h) + U_\mathrm{vdW}(h) = U_0\,\mathrm{e}^{-\kappa h} - \frac{A}{12\pi h^2}.$$

The competition between a screened repulsion (range λ_D) and an unscreened power-law attraction sets the stability.

6.3 Essential Results and How They Arise

Van der Waals to Hamaker. Integrating the atomic $-L/r^6$ dispersion attraction over two semi-infinite plates raises the power of the distance law twice ($r^{-6} \to D^{-3} \to h^{-2}$), giving $U_\mathrm{vdW} = -A/12\pi h^2$ with $A = \pi^2 n^2 L$. The macroscopic force is far longer-ranged than the atomic one, and A packages all the material-specific physics into one number ($\sim 10^{-20}$ J).

Debye–Hückel screening. Combining Poisson's equation with the Boltzmann ion distribution gives the nonlinear Poisson–Boltzmann equation; linearising for $e\phi \ll k_\mathrm{B}T$ yields $\nabla^2\phi = \kappa^2\phi$, so the potential decays as $e^{-\kappa x}$. The screening length $\lambda_D = 1/\kappa$ *shrinks* as salt is added ($\kappa^2 \propto \sum z_i^2 n_{i0}$), which is the key control knob for colloidal stability.

DLVO and stability. Adding the screened repulsion to the van der Waals attraction produces a rich potential: a deep primary minimum at contact (aggregation), often a repulsive barrier at intermediate range (kinetic stability), and sometimes a shallow secondary minimum at larger range (weak, reversible flocculation). High surface charge and low salt give a tall barrier (stable suspension); adding salt lowers λ_D, shrinks the barrier, and triggers aggregation — the basis of "salting out".

Steric stabilisation. When charge stabilisation fails (e.g. in deionised water or at high salt), grafting hydrophilic polymer brushes provides an alternative: overlapping brushes raise the local osmotic pressure and cost conformational entropy, both repulsive and essentially independent of ionic strength (provided the solvent stays good, $\chi < \frac{1}{2}$).

6.4 Diagram Analysis

The DLVO energy curves are the centrepiece: plotting $U(h)$ shows how the total potential shifts from purely attractive (unstable, low charge or high salt) to developing a repulsive barrier (stable) as surface charge rises or salt falls. The position of the secondary minimum often coincides with optical wavelengths, giving rise to Bragg-scattering "photonic" colloidal crystals. The Poisson–Boltzmann potential plots show the linear Debye–Hückel decay holding at low surface charge but deviating near a highly charged surface, where counter-ion enrichment (unbounded) outpaces co-ion depletion (bounded at zero).

Key ideas & pitfalls

1. Bjerrum vs Debye length. The Bjerrum length l_B (fixed by the solvent, ~ 0.7 nm in water) is the range of the *bare* charge–charge interaction at strength $k_{\mathrm{B}}T$; the Debye length λ_D (tunable by salt) is the range of the *screened* interaction. Don't conflate them.
2. Salt destabilises charge-stabilised colloids. More salt $\Rightarrow$ shorter $\lambda_D \Rightarrow$ weaker repulsion $\Rightarrow$ aggregation. Steric stabilisation, by contrast, is immune.
Takeaway: colloidal stability is a competition of two interactions — unscreened van der Waals attraction versus screened electrostatic (or entropic steric) repulsion — and the whole of DLVO is reading off who wins as a function of separation.

7 Self-Assembly: Surfactants and Membranes

7.1 Key Concepts

- **Amphiphile / surfactant:** a molecule with a hydrophilic head and hydrophobic tail; the dual character drives self-assembly.
- **Hydrophobic effect:** the entropic penalty of ordering water around a hydrophobic tail, which is what actually drives aggregation.
- **Critical micelle concentration (CMC):** the concentration above which added surfactant forms micelles rather than free monomers.
- **Packing parameter $p = v/a_0\ell$:** the geometric ratio predicting aggregate shape (sphere, cylinder, bilayer).
- **Self-assembly thermodynamics:** the law of mass action for aggregation number N, giving the sharp onset at the CMC.
- **Membrane elasticity:** bending rigidity κ and tension σ; the Helfrich energy.

- **Membrane fluctuation spectrum:** thermal undulations with $\langle |h_q|^2 \rangle \propto k_\mathrm{B}T/(\sigma q^2 + \kappa q^4)$.

7.2 Core Equations

Packing parameter and aggregate shape

$$p = \frac{v}{a_0 \ell}: \quad p < \tfrac{1}{3} \text{ spheres, } \tfrac{1}{3} < p < \tfrac{1}{2} \text{ cylinders, } \tfrac{1}{2} < p < 1 \text{ bilayers.}$$

v is the tail volume (m^3), a_0 the head-group area (m^2), ℓ the tail length (m).

Aggregation and membrane fluctuations

$$X_A \text{ (monomer)} \leq (NK)^{-1/N}, \qquad \langle |h_{\vec{q}}|^2 \rangle = \frac{k_\mathrm{B}T}{A} \frac{1}{\sigma q^2 + \kappa q^4}.$$

X_A is the monomer mole fraction, K the equilibrium constant, N the aggregation number; $h_{\vec{q}}$ is the Fourier amplitude of the membrane height, κ the bending rigidity (J), σ the tension ($\mathrm{N\,m^{-1}}$), A the membrane area.

7.3 Essential Results and How They Arise

The hydrophobic effect drives assembly. Dissolving a hydrophobic tail forces surrounding water into an ordered cage, lowering its entropy. Hiding the tails in a micelle core releases that water, raising the entropy — so aggregation is *entropically* driven, not by tail–tail attraction. This is why it strengthens with temperature.

The CMC from mass action. For the equilibrium N monomers $\rightleftharpoons$ aggregate, the law of mass action with $K \gg 1$ and $N \gg 1$ shows the free-monomer concentration cannot exceed $(NK)^{-1/N}$ — a sharp ceiling. Add more surfactant and it goes into micelles, not monomers. That ceiling is the CMC.

Packing parameter sets shape. Geometry alone: a spherical micelle of α surfactants needs $R = 3v/a_0$, but the radius cannot exceed the tail length ℓ, giving $p = v/a_0\ell < \frac{1}{3}$. Repeating for cylinders and bilayers gives the $\frac{1}{2}$ and 1 bounds. Cone-shaped molecules (small p, bulky head) make high-curvature spheres; cylindrical molecules ($p \approx 1$) make flat bilayers — which is why double-tailed lipids form cell membranes.

Membrane fluctuations. Expanding the Helfrich energy $E = \int [\frac{\sigma}{2}(\nabla h)^2 + \frac{\kappa}{2}(\nabla^2 h)^2]$ in Fourier modes diagonalises it into independent harmonic modes; equipartition ($\frac{1}{2}k_\mathrm{B}T$ each) gives $\langle |h_q|^2 \rangle = k_\mathrm{B}T/A(\sigma q^2 + \kappa q^4)$. Tension dominates long wavelengths ($q < \sqrt{\sigma/\kappa}$), bending dominates short ones.

7.4 Diagram Analysis

The packing-parameter figure links molecular shape to aggregate morphology: cone $\to$ sphere, truncated cone $\to$ cylinder/vesicle, cylinder $\to$ planar bilayer, inverted cone $\to$ inverted phases — a single geometric ratio predicting the whole sequence. The surface-tension -versus-concentration plot shows γ falling linearly in $\ln c$ until the CMC, then flattening. The membrane-fluctuation sketches (flicker spectroscopy on red blood cells) show how measuring the undulation spectrum extracts the bending rigidity κ.

Key ideas & pitfalls

1. Self-assembly is entropy-driven. The hydrophobic effect is about *water's* entropy, not an attraction between tails. This is why micellisation is endothermic yet spontaneous.
2. The packing parameter is purely geometric. It predicts shape from three single-molecule measurements with no energetics — a remarkably powerful zero-parameter argument.
Takeaway: amphiphiles self-organise because hiding their tails from water lowers the global free energy; geometry (the packing parameter) then dictates the shape, and the same equipartition-over-modes machinery from earlier chapters gives the membrane's thermal undulations.

8 Synthesis: The Threads That Run Through the Course

Recurring ideas

- **Energies of order $k_\mathrm{B}T$.** Every structure — a stretched chain, a screening layer, a micelle, an undulating membrane — is held together by energies comparable to thermal energy, so all of them fluctuate and all are easily deformed.
- **Entropy as a force.** The entropic spring (polymers), the osmotic repulsion of brushes (colloids), the hydrophobic effect (self-assembly), and the entropy of mixing (Flory–Huggins) are all entropic forces with no energetic counterpart.
- **Equipartition over modes.** The same $\frac{1}{2}k_\mathrm{B}T$-per-mode counting gives Brownian displacement, capillary-wave roughness, and the membrane fluctuation spectrum.
- **Fluctuation and dissipation are linked.** The random thermal force and the systematic drag are two faces of the same molecular collisions (Stokes–Einstein), a theme that recurs whenever dynamics appear.
- **Overdamped, low-Reynolds dynamics.** Inertia is negligible throughout; motion is force-balance between driving and viscous drag, from swimming microorganisms to reptating DNA.
- **Competition of interactions.** Stability and morphology are decided by competing terms — attraction vs repulsion (DLVO), entropy vs energy (coil–globule, mixing), curvature vs tension (membranes).

8.1 Master Equation Reference

Topic	**Key relation**
Reynolds number	$Re = \rho v_0 L_0/\eta$; $Re \ll 1 \Rightarrow$ Stokes flow $0 = -\nabla p + \eta\nabla^2\vec{v}$
Stokes drag	$F = 6\pi\eta a v$
Poiseuille	$Q = \pi R^4 \Delta p/8\eta L$, $R_{\text{hyd}} = 8\eta L/\pi R^4$
Stokes–Einstein	$D = k_{\text{B}}T/\gamma = k_{\text{B}}T/6\pi\eta r$, $\langle x^2\rangle = 6Dt$
Viscoelastic models	Maxwell (series): stress relaxation $G\mathrm{e}^{-t/\tau}$; KV (parallel): bounded creep; Zener: both
Ideal chain	$\langle R^2\rangle = Nb_0^2$; spring $f = 3k_{\text{B}}TR/Nb_0^2$
Real chain	$R \sim bN^{3/5}$ (good solvent), $N^{1/2}$ (theta), $N^{1/3}$ (globule)
Flory–Huggins	$f/k_{\text{B}}T = \frac{\Phi}{N}\ln\Phi + (1-\Phi)\ln(1-\Phi) + \chi\Phi(1-\Phi)$; $\chi_c \approx \frac{1}{2} + 1/\sqrt{N}$
Rubber elasticity	$\sigma = \nu_s k_{\text{B}}T(\lambda - \lambda^{-2})$, $G = \nu_s k_{\text{B}}T$
Surface tension	$\Delta p = \gamma(1/R_1 + 1/R_2)$; Jurin $h = 2\gamma\cos\Theta/\rho g R$
Young	$\gamma_{SL} + \gamma_{LV}\cos\Theta = \gamma_{SV}$
Bjerrum / Debye	$l_B = e^2/4\pi\epsilon_0\epsilon_r k_{\text{B}}T$; $\kappa^2 = e^2\sum_i z_i^2 n_{i0}/\epsilon k_{\text{B}}T$
Van der Waals	$U_{\text{vdW}} = -A/12\pi h^2$, $A = \pi^2 n^2 L$
DLVO	$U = U_0\mathrm{e}^{-\kappa h} - A/12\pi h^2$
Packing parameter	$p = v/a_0\ell$: sphere $< \frac{1}{3} <$ cylinder $< \frac{1}{2} <$ bilayer
Membrane	$\langle \lvert h_q\rvert^2\rangle = k_{\text{B}}T/A(\sigma q^2 + \kappa q^4)$
Electrophoresis	$u_e = \epsilon_r\epsilon_0\zeta/\eta$ (Helmholtz–Smoluchowski; appendix)

8.2 Final Word

The intellectual economy of soft matter is that a few ideas — thermal energy as the relevant scale, entropy as a force, equipartition over fluctuating modes, overdamped dynamics, and the competition of weak interactions — recur in every chapter. A polymer coil, a colloidal suspension, and a lipid membrane look entirely different, but each is a structure of characteristic energy $\sim k_{\text{B}}T$ whose behaviour is read off from the same handful of statistical-mechanical principles. Identify which free energy is competing with which, set $\partial F = 0$ or invoke equipartition, and the physics follows.

End of summary. Read alongside the problem sheets and the Key Derivations companion, which works the central derivations through in full. Notation follows the course lecture notes; the electrokinetics material (electrophoresis, the Helmholtz–Smoluchowski equation) is drawn from the appendix, which is background rather than core examinable content.